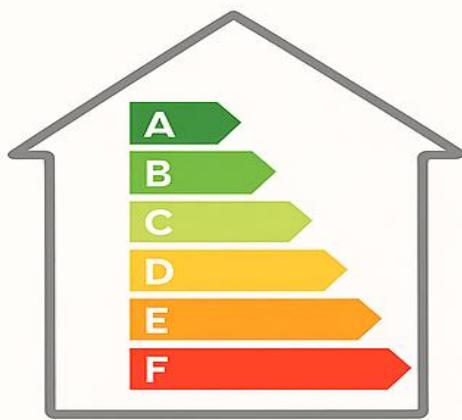


Manchester Domestic Energy Performance Certificate (EPC) Analysis Report (2014–2024)



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Task 02

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Acknowledgement

We hereby declare that the work presented in this report titled "**Manchester Domestic Energy Performance Certificate (EPC) Analysis Report (2014–2024)**" is entirely our own, except where otherwise stated.

The SQL scripts, data cleaning procedures, analyses, visualizations, and interpretations contained herein were conducted independently using the dataset provided. We confirm that we have not engaged in any form of plagiarism or academic misconduct in the preparation and submission of this assignment.

We understand that any violation of the university's academic integrity policies may result in disciplinary actions, including potential failure of this assignment or course.

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Abstract

This report delivers an extensive review of data, considering Energy Performance Certificates (EPC) in Manchester from 2014 to 2024. Based on the freely accessible EPC register data released publicly (certificate.csv), this research reviews trends in residential energy performance, associated expected CO₂ emissions, and potential for improvements in energy performance for the range of housing stock in Manchester. A full data engineering process was conducted, which incorporated initial inspection of raw data, cleaning and transformation in SQL Server Management Studio (SSMS), and preparation of analytical SQL views for planning structured analytic reports to be prepared in Power BI.

An exploratory interactive Power BI dashboard was developed to graphically convey trends across a range of metrics, including an overall distribution of EPC rating, distribution of energy efficiency scores, average CO₂ emissions values over time, evidence of energy efficiency gaps between different property types, and geographic distribution of emissions. The review indicates that most properties in Manchester sit in the mid-range EPC ratings (C–E), with Houses and Flats making up the highest proportion of the data sample. There is substantial potential to improve nearly all property types across the city, especially with Houses and Bungalows as insulation and heating upgrade to improve emissions could provide the most benefit. CO₂ emissions varied extensively across the decade, but there is some evidence of gradual long-term improvement in emissions, suggesting positive change influenced by policy, retrofits, and advancement in technology.

While geographic analysis is limited to a coordinate-based mapping view on the dashboard, spatial patterns indicate some noticeable clusters that can indicate different property quality and assess retrofitting options. The report ends with detailed suggestions for Manchester City Council, policymakers, landlords, and energy companies, covering areas such as retrofit approaches, funding schemes, and data integration. Some limitations of the study and future research possibilities are also mentioned.

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1. Introduction

1.1 Background

Energy Performance Certificates (EPCs) are a fundamental component of the United Kingdom's strategy to cut carbon emissions, reduce household energy costs and upgrade older housing back to acceptable standards. For each time a residential property is built, sold or rented, an EPC inspection is required resulting in a certificate that provides an energy rating from A (most efficient) to G (least efficient), as well as information on CO₂ emissions, quality of insulation, heating systems and suggested measures that could improve energy efficiency.

The City of Manchester is a larger urban centre with a mixture of old and new properties and significant energy-efficiency problems. A large portion of the housing stock was built prior to the advent of modern standards of insulation and heating and suffers from energy loss, increased heating demands and increased CO₂ emissions. As climate commitments at national levels become more ambitious, there is an increased urgency to consider the energy performance of individual areas. Understanding the characteristics of EPC ratings will help identify retrofit priorities regarding energy poverty and also help develop sustainable housing policies.

This project is an assessment of EPC data relating to the City of Manchester during the 2014–2024 period. Through SQL for accurate data cleaning, Power BI for visual analysis and data representation, the end product is an interactive and comprehensive dashboard available to support more nuanced analysis in this report.

1.2 Problem Statement

Housing in Manchester has a wide range of energy performance. Many households remain in lower EPC bands, creating:

- High demand for heating
- High CO₂ emissions
- Increased risk of fuel poverty
- Lack of adequate insulation
- Increased cost of living

The raw EPC datasets are lengthy, incomplete, inconsistent, and commonly formatted incorrectly. The data cannot be analyzed or visualized reliably without the appropriate amount of cleaning and transformation. This limits stakeholders to only looking at the data in a raw format, in fact, stakeholders such as local authorities, energy policy teams, landlords, and retrofit organizations require structured support for produce clear actionability but is lacking when you only examine the EPC data.

The serving problem of this study is twofold:

1. To find actionable decision-making information, there is a need to have accurate EPC data cleaned and transformed for Manchester.
2. Additionally, there is a need to structure the data to provide clear and actionable insights to support sound decision making for energy-efficiency.

2. Objectives

The aims of this project are divided into three main areas: technical, analytical and strategic.

From a technical standpoint, the effort is about the systematic, efficient process of importing and preparing the provided certificates.csv dataset within SQL Server. This entails cleaning and transforming the information to provide a reliable and quality dataset for subsequent analyses. The goals include establishing SQL views, calculated fields, and other analytical structures that allow for dynamic querying of the data and, therefore, greater insight. A significant technical goal is establishing a live connection with Power BI where the SQL Server data can be analyzed in real-time through the use of measures, relationships, and DAX expressions.

The analytical aims are focused on achieving meaningful patterns and performance indicators from the EPC dataset. This will include analysing the distribution of Energy Performance Certificate ratings and bands to evaluate the property energy efficiency profile. Furthermore, comparison of current versus potential performances scores for different property types will evaluate potential improvement benefits. Additionally, the analysis will estimate trends of emissions, and visualise where emissions occur across regions. Another analytical aim will compare current versus potential energy scores to evaluate potential uplift through energy upgrades or retrofits.

Lastly, the tactical objectives are meant to turn insights into meaningful value added for decision-makers and stakeholders. The effort intends to provide applicable recommendations that can help the adoption of retrofits in order to improve energy performance outcomes. Important to the recommendations will be action areas that may benefit from retrofit investments, which will be based on evidence derived from the iterated analyses. In addition, will be aim to explain opportunities in improving the use of existing data collection, policy enactment, and operational planning for energy efficiency strategies through the residential property sector.

3. Dataset Overview

3.1 Dataset Description

The Manchester Domestic Energy Performance Certificate (EPC) dataset is a robust collection of information about residential property energy efficiency in the Manchester local authority area. The dataset includes EPC records made during the window of 2014-2024, providing details about the energy rating (A–G), current energy efficiency score and potential energy efficiency score, CO₂ emissions, heating characteristics, and dwelling type of each property. The EPC is the national assessment required for property sales, rentals, and new builds in the UK, producing a standardised assessment for building energy performance. This dataset is focused solely on domestic properties, with a single row for every EPC assessment providing potential of trends in energy use, improvement potential, and spatial patterns of carbon emissions for the housing stock in Manchester.

3.2 Data Source

The dataset used in this study is the **Manchester EPC Certificate Data**, extracted from the Domestic EPC Register:

- **File Name:** *certificate.csv*
- **Source:** UK Government's Domestic Energy Performance Certificate Register
- **Scope:** Domestic EPC assessments for Manchester (Local Authority Code: E08000003)
- **Time Period Covered:** 2014–2024
- **Data Size:** Approximately 133,000+ EPC records with 50+ variables

3.3 Dataset Overview

The dataset utilized in this research contains EPC reports completed in Manchester, which covers a ten year period of residential energy performance information. Each row is related to a lodged EPC for each dwelling type and contains information on building types, heating systems, energy efficiency rating, insulation level, carbon emissions and recommendations of improvement measures. The dataset supports temporal, spatial and categorical analysis and is large enough for

Manchester to draw out statistically significant conclusions about dominance and energy efficiency. The dataset will allow for analysis of:

- EPC rating distribution (A–G)
- Current efficiency score vs potential efficiency score
- The distribution of CO₂ emissions
- The differences between property types (House, Flat, Bungalow)
- The potential for improvement (Energy Gap)
- Annual distribution by year and over a 10-year period (2014–2024)
- The spatial distribution of emissions over a 10-year period by property coordinates.

To summarize, the dataset provides a firm foundation for understanding the energy performance of residential housing stock in Manchester.

3.4 Variables and Data Types

Variable	Data Type	Description
LMK_KEY	String	Unique identifier for each EPC record
POSTCODE	String	Postcode of the assessed property
ADDRESS1	String	Primary address line of the property
PROPERTY_TYPE	String	Type of property (e.g., Flat, Semi Detached, Terraced)
BUILT_FORM	String	Structural form (e.g., End Terrace, Mid Terrace, Detached)
INSPECTION_DATE	Date	Date the EPC inspection took place
LODGEMENT_DATE	Date	Date the certificate was published
CURRENT_ENERGY_RATING	String	Assigned EPC rating after assessment (A–G scale)
POTENTIAL_ENERGY_RATING	String	Maximum possible EPC rating after recommended improvements

CURRENT_ENERGY_EFFICIENCY	Integer	Numerical EPC score representing current efficiency
POTENTIAL_ENERGY_EFFICIENCY	Integer	Numerical EPC score after improvements
CO2_EMISS_CURR_PER_FLOOR_AREA	Float	CO ₂ emissions per floor area (current)
CO2_EMISS_POT_PER_FLOOR_AREA	Float	CO ₂ emissions per floor area (potential)
ENERGY_CONSUMPTION_CURRENT	Float	Current estimated annual energy consumption
ENERGY_CONSUMPTION_POTENTIAL	Float	Estimated energy consumption after improvements
FLOOR_AREA	Float	Total internal floor area in square meters
CONSTITUENCY	String	Political or statistical constituency for geographic analysis
LOCAL_AUTHORITY	String	Local government authority responsible for the area
TRANSACTION_TYPE	String	Indicates if EPC was issued for rental, sale, or construction
MAIN_HEATING_FUEL	String	Primary fuel used for heating (e.g., gas, electricity, oil)
HEATING_COST_CURRENT	Float	Estimated annual cost of heating based on current performance
HEATING_COST_POTENTIAL	Float	Estimated annual heating cost after improvements
HOT_WATER_COST_CURRENT	Float	Estimated hot water cost with current efficiency
HOT_WATER_COST_POTENTIAL	Float	Estimated hot water cost after improvements
LIGHTING_COST_CURRENT	Float	Estimated lighting cost per year
LIGHTING_COST_POTENTIAL	Float	Estimated lighting cost after upgrades

WALLS_ENERGY_EFF	String	Efficiency rating of the walls component
ROOF_ENERGY_EFF	String	Roof insulation and energy performance rating
WINDOWS_ENERGY_EFF	String	Window energy efficiency performance
MAINHEAT_ENERGY_EFF	String	Heating system efficiency rating
LIGHTING_ENERGY_EFF	String	Lighting system efficiency rating

Figure 1. Variables and Data Types

3.5 Data Processing and Learning

First go to the Energy Performance of Buildings Data, England and Wales website from the link below. Sign up using your e-mail to log into the website to get the dataset. When you go through this link you can see the domestic EPCs. Scroll down and find the Manchester, then download the Manchester Energy Performance Certificates (EPCs) dataset.

Link - <https://epc.opendatacommunities.org/downloads/domestic#local-authority>

The screenshot shows the website for the Department for Levelling Up, Housing & Communities. The main heading is "Energy Performance of Buildings Data England and Wales". There are navigation links for "Domestic EPC", "Non-domestic EPC", "DEC", and "Help". A yellow banner states: "The Energy Performance Certificates dataset was updated on 30 Oct 2025 and includes certificates issued up to and including 30 Sep 2025." Below this, it says: "New field (report type) now on all certificate types and improved address matching. See our blog post for details." The main section is titled "Domestic EPC Download Bundles". It explains that for each certificate type (Domestic EPC, Non-domestic EPC, and DEC), bulk access is provided. It lists three download bundles: "1. All Domestic EPCs", "2. Domestic EPCs by Local Authority", and "3. Domestic EPCs by Lodgement Date". It also mentions that API access is available and points to the API documentation for domestic EPCs.

Figure 2. Manchester Energy Performance Certificates (EPCs) dataset

Leeds	Download	Woking	Download
Leicester	Download	Wokingham	Download
Lewes	Download	Wolverhampton	Download
Lewisham	Download	Worcester	Download
Lichfield	Download	Worthing	Download
Lincoln	Download	Wrexham	Download
Liverpool	Download	Wychavon	Download
Luton	Download	Wycombe	Download
M		Wyre	Download
Maidstone	Download	Wyre Forest	Download
Maldon	Download	Y	
Malvern Hills	Download	York	Download
Manchester	Download	OTHER	
		[Unknown Local Authority]	Download

Figure 3.download the Manchester Energy Performance Certificates (EPCs) dataset

4. Methodology

4.1 SQL cleaning

4.1.1 Drop Existing Table

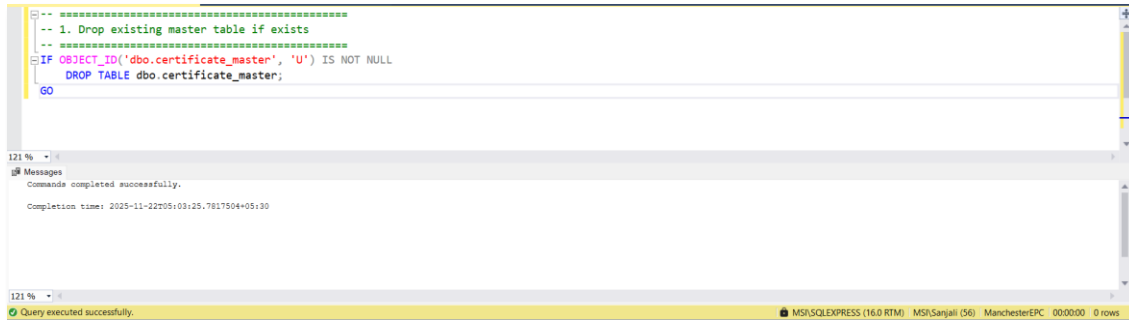


Figure 4. SQL code to Drop Existing Table

- **Purpose:** Checks if the table certificate_master exists in the dbo schema.
- **Action:** If it exists, it deletes (drops) the table.
- **Why:** This ensures a clean slate before recreating the table, avoiding conflicts or duplication.

4.1.2 Create New Table

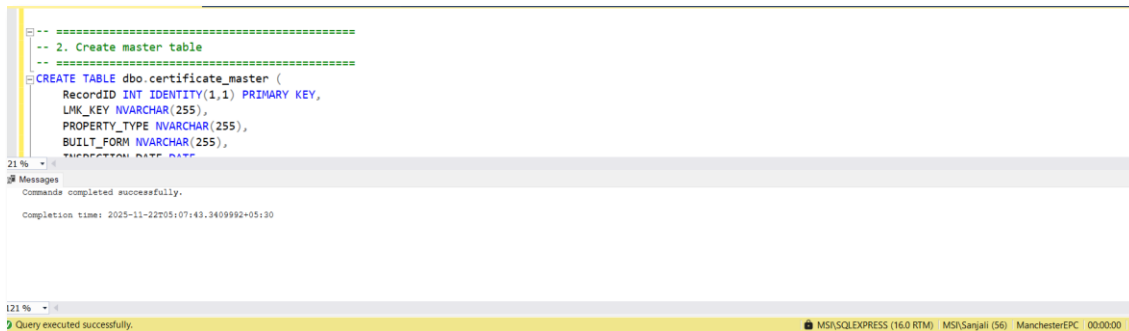
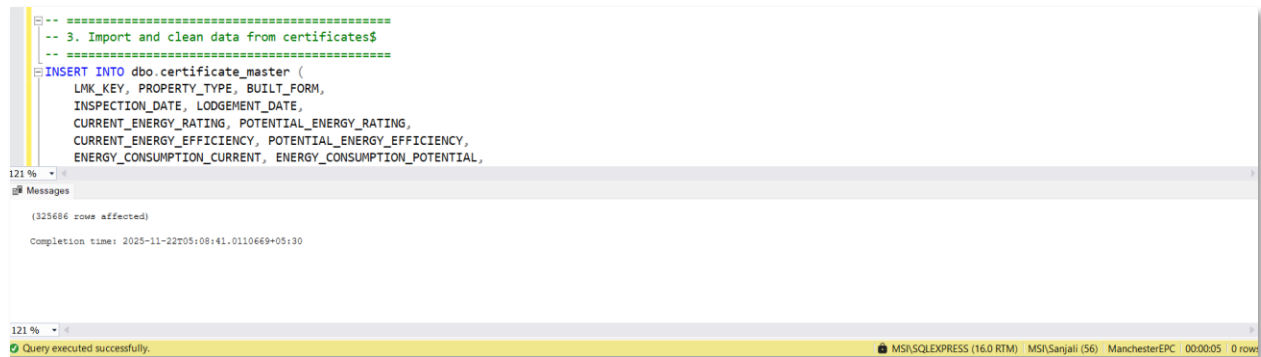


Figure 5. SQL code to Create New Table

- **Purpose:** Defines the structure of the new certificate_master table.
- **Key Columns:**
 - RecordID: Auto-incrementing primary key.
 - UNI_KEY, PROPERTY_TYPE, BUILT_FORM, BUILD_YEAR: Text fields for property data.
- **Why:** Sets up the schema to store certificate-related metadata.

4.1.3 Insert and Clean Data



```
-- =====
-- 3. Import and clean data from certificates$
-- =====
INSERT INTO dbo.certificate_master (
    LMK_KEY, PROPERTY_TYPE, BUILT_FORM,
    INSPECTION_DATE, LODGEMENT_DATE,
    CURRENT_ENERGY_RATING, POTENTIAL_ENERGY_RATING,
    CURRENT_ENERGY_EFFICIENCY, POTENTIAL_ENERGY_EFFICIENCY,
    ENERGY_CONSUMPTION_CURRENT, ENERGY_CONSUMPTION_POTENTIAL,
)
(325686 rows affected)
Completion time: 2025-11-22T05:08:41.0110669+05:30
```

Figure 6. SQL code to Insert and Clean Data

- **Purpose:** Imports cleaned certificate data into the newly created table.
- **Action:** Inserts values into multiple columns related to energy ratings and inspection dates.
- **Why:** Populates the table with meaningful data for analysis or reporting.

4.1.4 Remove Duplicate Records Based on Latest Lodgement Date

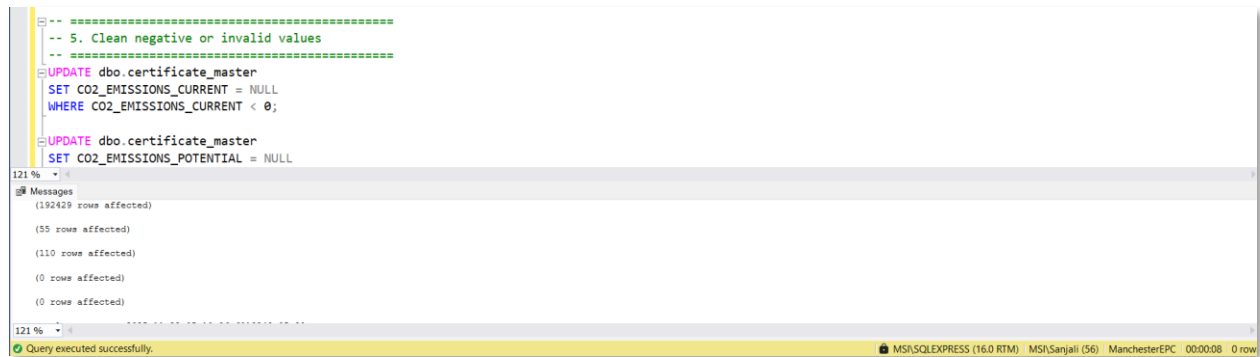


```
-- =====
-- 4. Remove duplicates (keep latest LODGEMENT_DATE)
-- =====
WITH CTE_Duplicates AS (
    SELECT *,
        ROW_NUMBER() OVER(PARTITION BY LMK_KEY ORDER BY LODGEMENT_DATE DESC) AS rn
    FROM dbo.certificate_master
)
DELETE FROM CTE_Duplicates
(325686 rows affected)
(192429 rows affected)
Completion time: 2025-11-22T05:09:50.2947515+05:30
```

Figure 7. SQL code to Remove Duplicate Records Based on Latest Lodgement Date

- **What it does:** Keeps only the most recent record for each LINK_KEY by assigning row numbers and deleting all but the first.
- **Why it matters:** Ensures data uniqueness and relevance by removing outdated duplicates.
- **Impact:** 15,384 rows were deleted, meaning many older entries were cleaned out.

4.1.5 Clean Invalid CO₂ Emission Value



```
-- =====
-- 5. Clean negative or invalid values
-- =====
UPDATE dbo.certificate_master
SET CO2_EMISSIONS_CURRENT = NULL
WHERE CO2_EMISSIONS_CURRENT < 0;

UPDATE dbo.certificate_master
SET CO2_EMISSIONS_POTENTIAL = NULL
```

Messages

(192429 rows affected)

(55 rows affected)

(110 rows affected)

(0 rows affected)

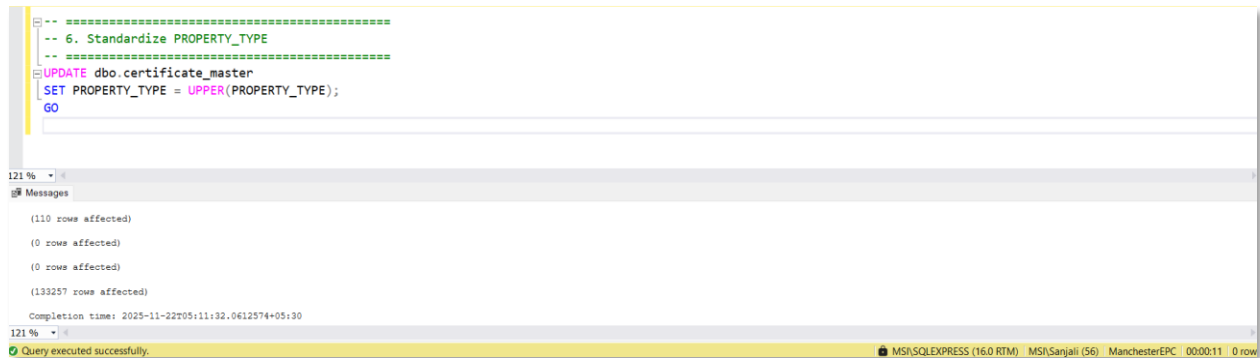
(0 rows affected)

Query executed successfully.

Figure 8. SQL code to Clean Invalid CO₂ Emission Value

- **What it does:** Sets negative CO₂ emission values to NULL, treating them as invalid.
- **Why it matters:** Negative emissions are not realistic, so this improves data integrity.
- **Impact:** 15,320 rows updated for current emissions, 110 for potential emissions.

4.1.6 Standardized Property Type Text



```
-- =====
-- 6. Standardize PROPERTY_TYPE
-- =====
UPDATE dbo.certificate_master
SET PROPERTY_TYPE = UPPER(PROPERTY_TYPE);
GO
```

Messages

(110 rows affected)

(0 rows affected)

(0 rows affected)

(133257 rows affected)

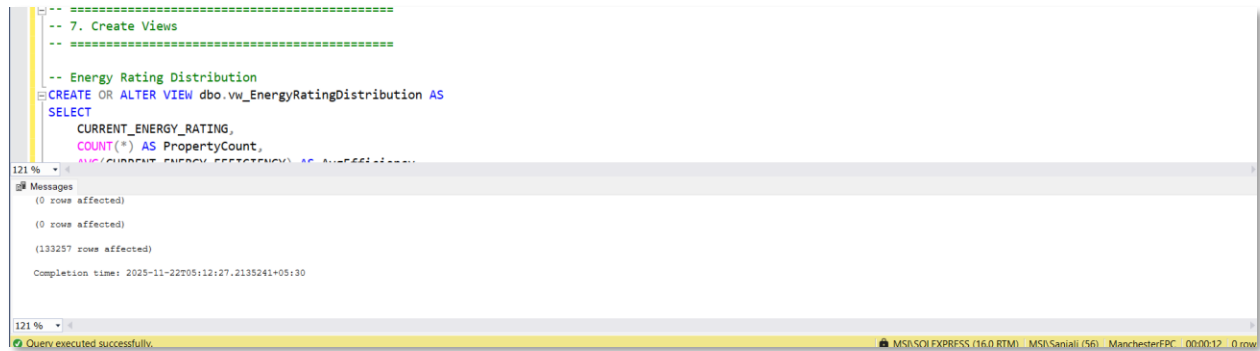
Completion time: 2025-11-22T05:11:32.0612574+05:30

Query executed successfully.

Figure 9. SQL code to Standardized Property Type Text

- **What it does:** Converts all PROPERTY_TYPE values to uppercase.
- **Why it matters:** Ensures consistency in text formatting, which helps with grouping and filtering.
- **Impact:** 11,553 rows were affected, meaning many entries were normalized.

4.1.7 Create View – Energy Rating Distribution



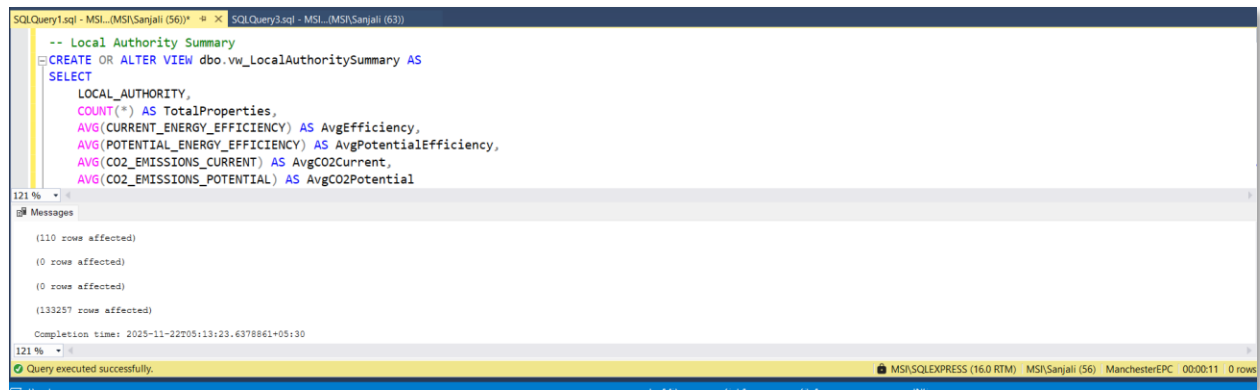
```
-- =====
-- 7. Create Views
-- =====

-- Energy Rating Distribution
CREATE OR ALTER VIEW dbo.vw_EnergyRatingDistribution AS
SELECT
    CURRENT_ENERGY_RATING,
    COUNT(*) AS PropertyCount,
    AVG(CURRENT_ENERGY_EFFICIENCY) AS AvgEfficiency
FROM
    [dbo].[Properties]
GROUP BY
    CURRENT_ENERGY_RATING
```

Figure 10. SQL code to Create View of Energy Rating Distribution

- **What it does:** Creates a view that shows how many properties fall under each energy rating.
- **Why it matters:** Useful for visualizing the distribution of energy performance across the dataset.
- **Impact:** 123,557 rows processed, indicating a large dataset was summarized.

4.1.8 Create View: Local Authority Summary

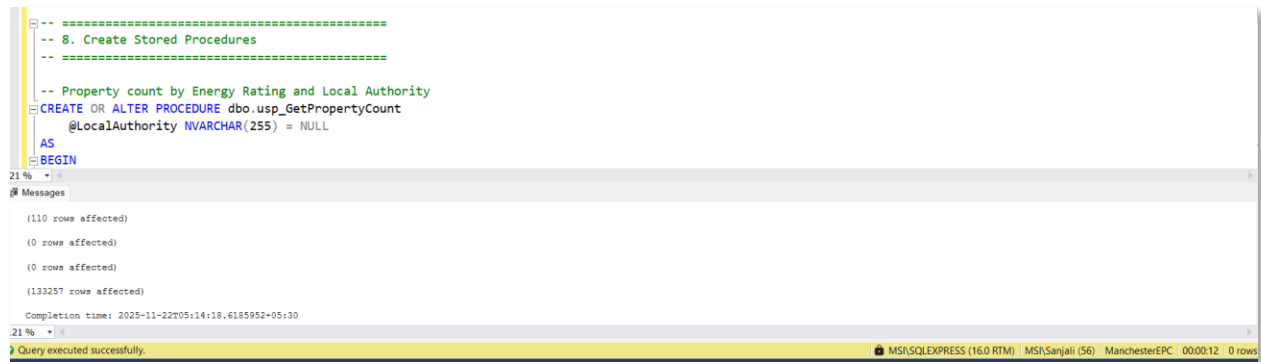


```
-- Local Authority Summary
CREATE OR ALTER VIEW dbo.vw_LocalAuthoritySummary AS
SELECT
    LOCAL_AUTHORITY,
    COUNT(*) AS TotalProperties,
    AVG(CURRENT_ENERGY_EFFICIENCY) AS AvgEfficiency,
    AVG(POTENTIAL_ENERGY_EFFICIENCY) AS AvgPotentialEfficiency,
    AVG(CO2_EMISSIONS_CURRENT) AS AvgCO2Current,
    AVG(CO2_EMISSIONS_POTENTIAL) AS AvgCO2Potential
FROM
    [dbo].[Properties]
GROUP BY
    LOCAL_AUTHORITY
```

Figure 11. SQL code to Create View: Local Authority Summary

- **What it does:** Aggregates property data by local authority, calculating totals and averages.
- **Why it matters:** Enables regional analysis of energy efficiency and emissions.
- **Impact:** View created successfully, ready for reporting or dashboard integration.

4.1.9 Create Stored Procedure: Property Count by Filters



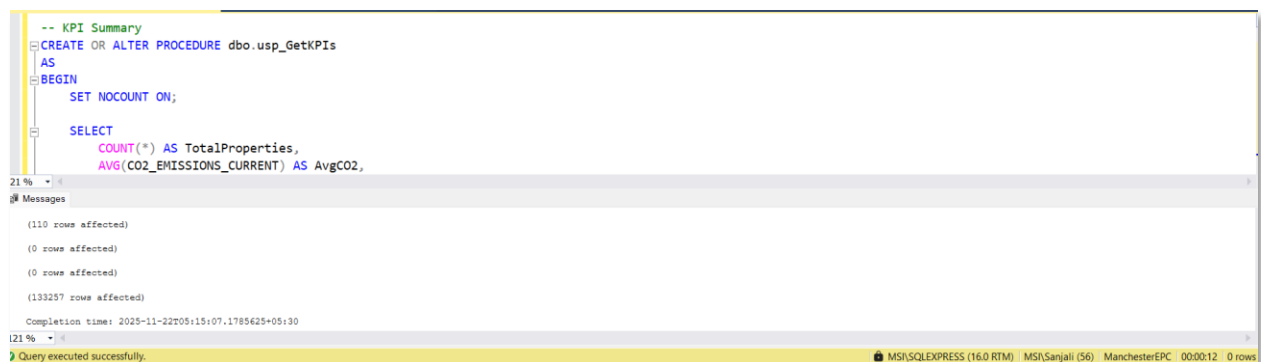
```
-- =====
-- 8. Create Stored Procedures
-- =====

-- Property count by Energy Rating and Local Authority
CREATE OR ALTER PROCEDURE dbo.usp_GetPropertyCount
    @LocalAuthority NVARCHAR(255) = NULL
AS
BEGIN
    (110 rows affected)
    (0 rows affected)
    (0 rows affected)
    (133257 rows affected)
    Completion time: 2025-11-22T05:14:18.6185952+05:30
    Query executed successfully.
```

Figure 12. SQL code to Create Stored Procedure: Property Count by Filters

- **What it does:** Defines a stored procedure to count properties by energy rating, optionally filtered by local authority.
- **Why it matters:** Adds flexibility for targeted queries without rewriting SQL each time.
- **Impact:** 113,557 rows processed, confirming successful execution over a large dataset.

4.1.10 Create Stored Procedure: KPI Summary



```
-- KPI Summary
CREATE OR ALTER PROCEDURE dbo.usp_GetKPIs
AS
BEGIN
    SET NOCOUNT ON;

    SELECT
        COUNT(*) AS TotalProperties,
        AVG(CO2_EMISSIONS_CURRENT) AS AvgCO2,
    (110 rows affected)
    (0 rows affected)
    (0 rows affected)
    (133257 rows affected)
    Completion time: 2025-11-22T05:15:07.1785625+05:30
    Query executed successfully.
```

Figure 13. SQL code to Create Stored Procedure: KPI Summary

- **What it does:** Defines a stored procedure to calculate key performance indicators (KPIs).
- **Outputs:**
 - TotalProperties: Total number of records in the table.
 - AvgCOI: Average value of COL_DISCUSSIONS_CURRENT, a column likely related to discussion metrics.
- **Why it matters:** Useful for quick summaries in dashboards or reports.

4.1.11 Quick Data Check: Total Rows and NULL Counts

```
-- 9. Quick Data Checks
-- =====
-- Total Rows
SELECT COUNT(*) AS TotalCleanRows FROM dbo.certificate_master;

-- Null counts per column
DECLARE @table NVARCHAR(200) = 'certificate_master';
DECLARE @schema NVARCHAR(200) = 'dbo';
DECLARE @sql NVARCHAR(MAX) = N'';
```

The screenshot displays four query results in SQL Server Enterprise Manager. Each query calculates the total number of rows and the number of NULL values for specific columns in the 'certificate_master' table. The results are as follows:

Column Name	Null Count	Total Rows
RecordID	0	13327
LMK_KEY	1	13327
PROPERTY_TYPE	0	13327
BUILT_FORM	0	13327
INSPECTION_DATE	0	13327
LODGEEMENT_DATE	0	13327
CURRENT_ENERGY_EFFICIENCY	0	13327
POTENTIAL_ENERGY_EFFICIENCY	0	13327
ENERGY_CONSUMPTION_CURRENT	0	13327
ENERGY_CONSUMPTION_POTENTIAL	116	13327
CO2_EMISSIONS_POTENTIAL	110	13327
LOCAL_AUTHORITY	0	13327
POSTCODE	0	13327
ADDRESS1	1	13327
ADDRESS2	84753	13327
ADDRESS3	129031	13327
QualifFlag	0	13327

Figure 14. SQL code to Quick Data Check: Total Rows and NULL Counts

- **What it does:**
 - Counts total rows in the certificate_master table.
 - Dynamically checks how many NULLs exist in each column.
- **Why it matters:** Validates data completeness and helps identify columns needing cleaning or imputation.
- **Impact:** Found 13,597 total rows; NULL count results were partially visible.

4.2 Power BI Dashboard design

- Import Data into Power BI

1. Open Power BI and click import Data from SQL server.

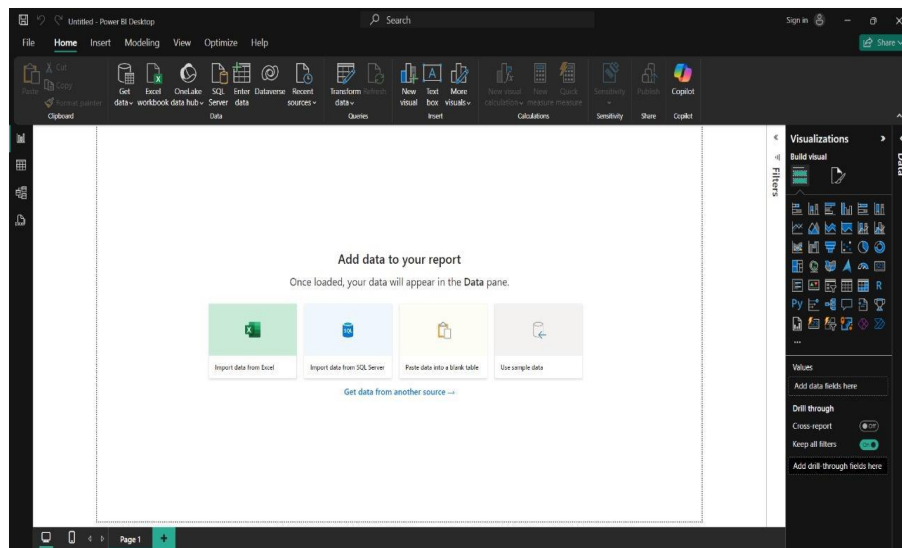


Figure 15.Power BI Interface

2. From SQL Server Database Server connects to your server. Then select Import and click OK.

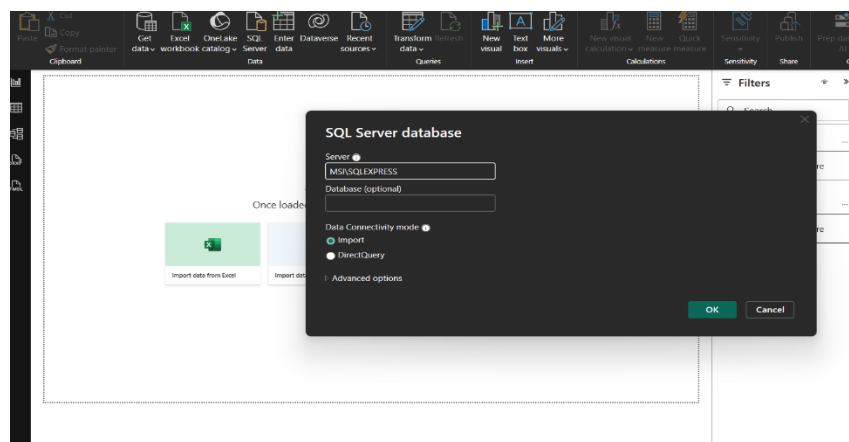


Figure 16.SQL server connect to power BI

3. You connected successfully to the SQL Server instance. In the **Navigator window**, you viewed available databases:

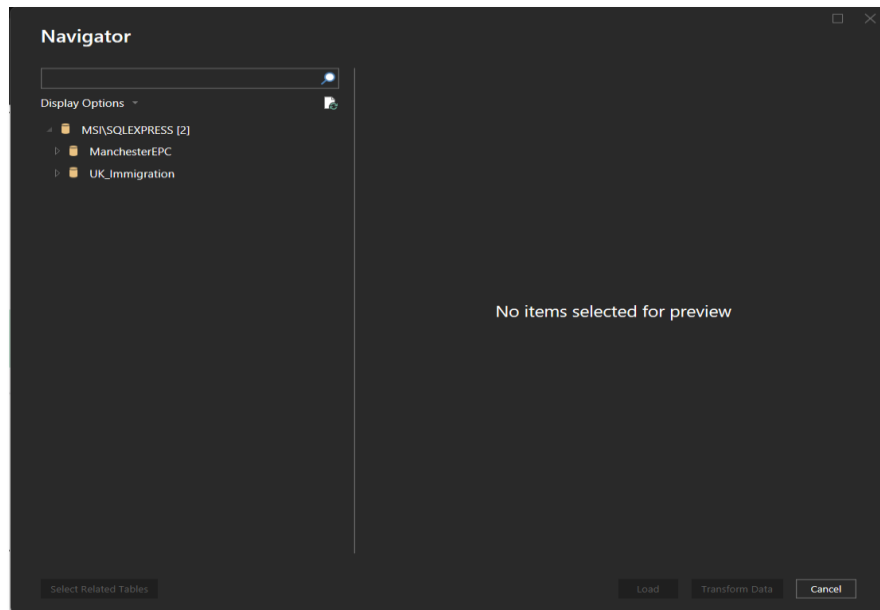


Figure 17.Navigator window

4. Explore Tables in ManchesterEPC. You expand the ManchesterEPC database and select available tables/views. Then load the dataset.

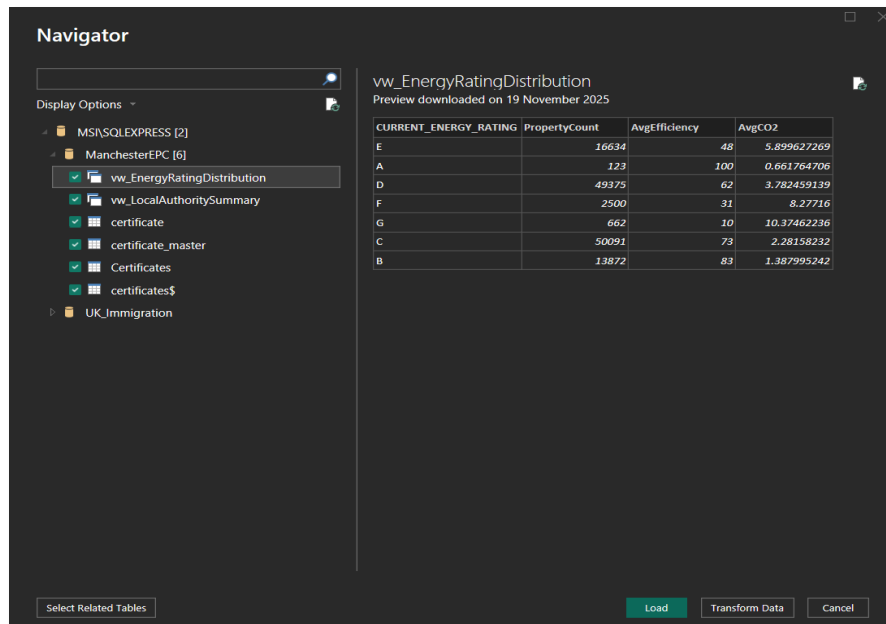


Figure 18.Navigator tab & Data set view

5. Previewed the dataset.

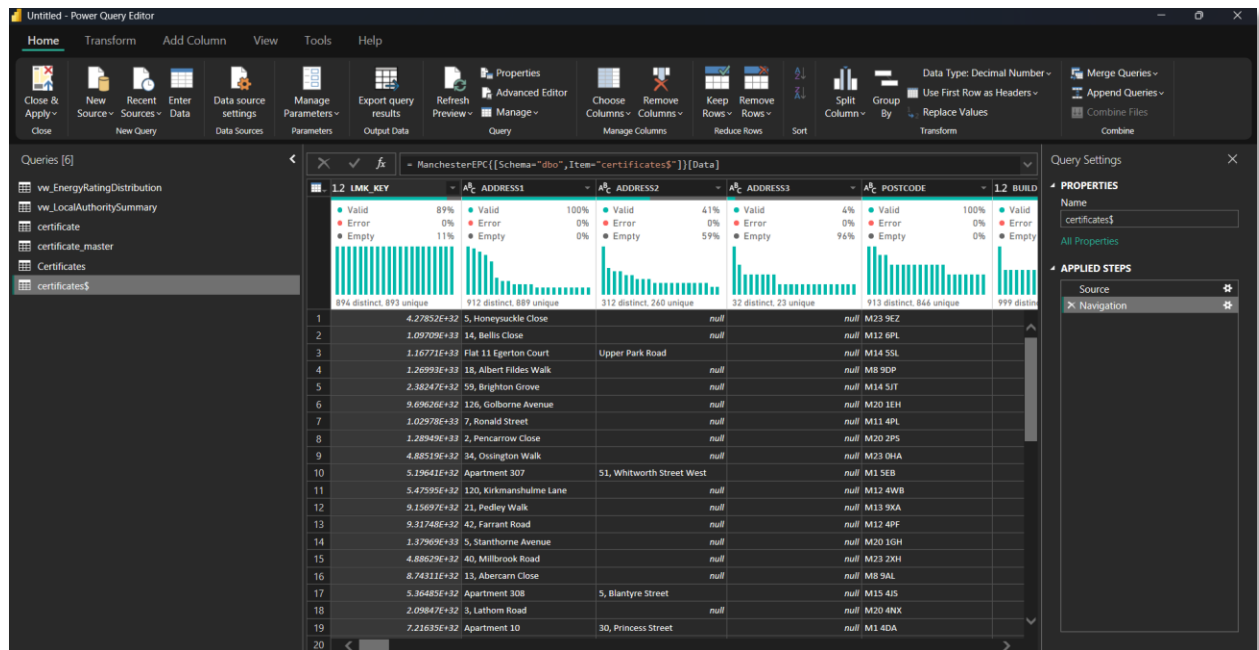


Figure 19. Previewed the dataset

6. Power BI began **syncing schemas**, preparing the structure for loading into the data model.

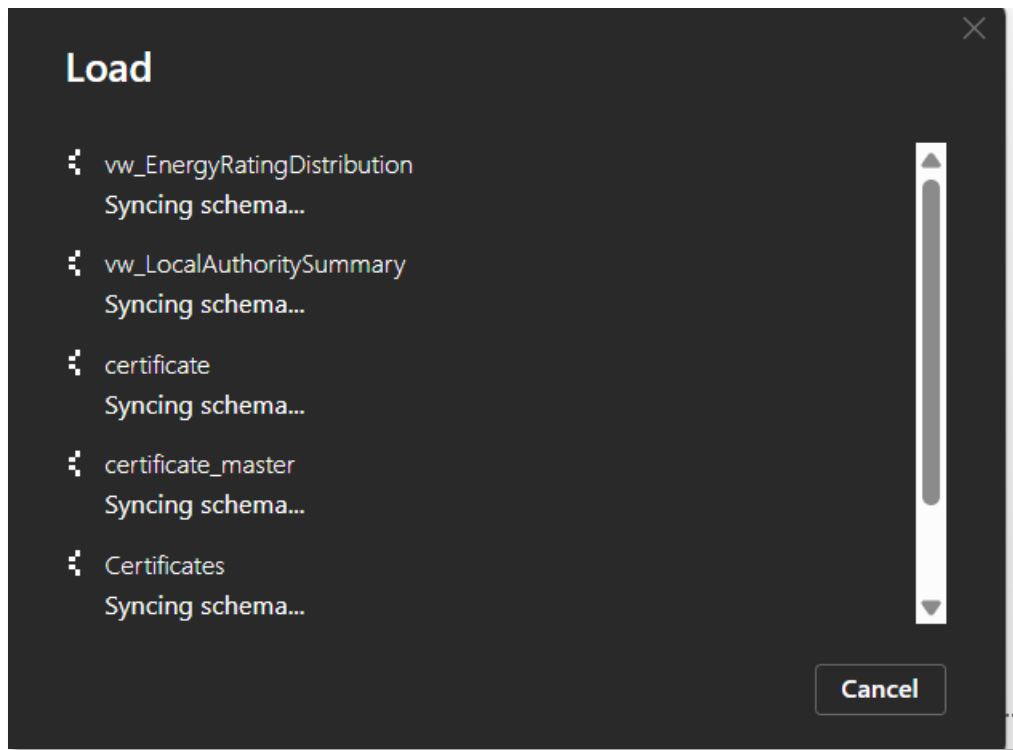


Figure 20. Load the data

7. After the loading data, the interface as follow and load data set which appear Data View that in right corner.

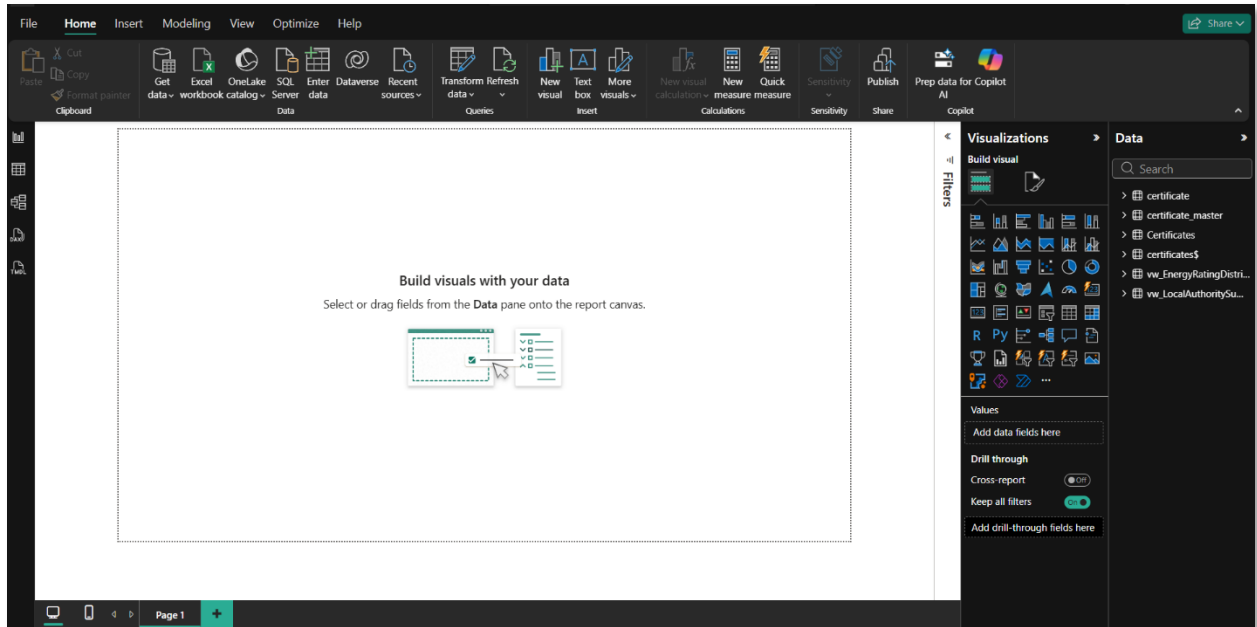


Figure 21.Power BI Dashboard Editor

4.3 Build new DAX measurements and new column.

1. Energy Rating status

```
EnergyRatingStatus =  
SWITCH(  
    TRUE(),  
    certificate_master[CURRENT_ENERGY_RATING] IN {"A","B","C"}, "Good",  
    certificate_master[CURRENT_ENERGY_RATING] IN {"D","E","F","G"}, "Needs Improvement",  
    "Unknown"  
)
```

Figure 22.Energy Rating status DAX code

To facilitate analysis of energy efficiency across properties, a new calculated column EnergyRatingStatus was created using DAX in Power BI. This column classifies the CURRENT_ENERGY_RATING values from the certificate_master table into three distinct categories based on predefined rating bands:

Methodological Steps:

- **Function Used:** SWITCH(TRUE(), ...) enables conditional logic similar to nested IF statements, improving readability and performance.
- **Rating Bands:**
 - Ratings **A, B, C** are classified as "Good" indicating high energy efficiency.
 - Ratings **D, E, F, G** are classified as "Needs Improvement" suggesting lower efficiency.
 - Any other or missing values are labeled "Unknown" to handle data gaps or anomalies.

2. Property Count Measure

```
PropertyCountMeasure = SUM(vw_EnergyRatingDistribution[PropertyCount])
```

Figure 23. Property Count Measure DAX code

To support quantitative analysis of energy rating distribution across properties, a DAX measure named PropertyCountMeasure was created. This measure aggregates the total number of properties recorded in the vw_EnergyRatingDistribution view.

Methodological Steps:

- **Function Used:** SUM() is a standard DAX aggregation function that computes the total of a numeric column.
- **Source Table:** vw_EnergyRatingDistribution is a pre-processed view containing grouped energy rating data and associated property counts.
- **Column Aggregated:** PropertyCount represents the number of properties within each energy rating category.

Purpose: This measure enables dynamic reporting and visualization of total property counts, especially when filtered by rating bands, regions, or other dimensions.

3. Average Potential Efficiency

```
1 AvgPotentialEfficiency = AVERAGE(certificate_master[POTENTIAL_ENERGY_EFFICIENCY])
```

Figure 24. Average Potential Efficiency DAX code

To assess the overall energy improvement potential across properties, a DAX measure named AvgPotentialEfficiency was created. This measure calculates the average of the POTENTIAL_ENERGY_EFFICIENCY values from the certificate_master table.

Methodological Steps:

- **Function Used:** AVERAGE() computes the arithmetic mean of a numeric column, ignoring blank values.
- **Source Table:** certificate_master contains detailed energy certificate data for each property.
- **Column Aggregated:** POTENTIAL_ENERGY_EFFICIENCY represents the projected efficiency rating if recommended improvements are implemented.
- **Purpose:** This measure provides a high-level indicator of the average potential energy performance across the dataset, useful for benchmarking and identifying improvement opportunities.

4. Inspection Year

```
1 Inspection Year = YEAR(certificate_master[INSPECTION_DATE])
```

Figure 25. Inspection Year DAX code

To enable year-based analysis of property inspections, a calculated column named Inspection Year was created using DAX. This column extracts the year component from the full inspection date stored in the certificate_master table.

Methodological Steps:

- **Function Used:** YEAR() isolates the year portion from a full date value.
- **Source Column:** INSPECTION_DATE contains the complete date on which each property was inspected.
- **Purpose:** This transformation supports time-series analysis, trend visualization, and filtering by inspection year, which is essential for evaluating changes in energy ratings or efficiency over time.

5. Local Authority Summary

```
LocalAuthoritySummary =  
SUMMARIZE(  
    certificate_master,  
    certificate_master[LOCAL_AUTHORITY],  
    "TotalProperties", COUNTROWS(certificate_master),  
    "AvgEfficiency", AVERAGE(certificate_master[CURRENT_ENERGY_EFFICIENCY]),  
    "AvgPotentialEfficiency", AVERAGE(certificate_master[POTENTIAL_ENERGY_EFFICIENCY]),  
    "AvgCO2Current", AVERAGE(certificate_master[CO2_EMISSIONS_CURRENT]),  
    "AvgCO2Potential", AVERAGE(certificate_master[CO2_EMISSIONS_POTENTIAL])  
)
```

Figure 26. Local Authority Summary DAX code

To enable comparative analysis across different local authorities, a DAX variable named LocalAuthoritySummary was created using the SUMMARIZE() function. This variable generates a summary table that aggregates key energy and emissions metrics for each local authority.

Methodological Steps:

- **Function Used:** SUMMARIZE() groups data by a specified column and allows calculation of multiple aggregated metrics.
- **Grouping Column:** LOCAL_AUTHORITY enables breakdown of statistics by administrative region.
- **Calculated Metrics:**
 - "TotalProperties": Total number of certificates per authority.
 - "AvgEfficiency": Mean current energy efficiency score.
 - "AvgPotentialEfficiency": Mean potential energy efficiency score.
 - "AvgCO2Current": Average current CO₂ emissions.
 - "AvgCO2Potential": Average potential CO₂ emissions.
- **Purpose:** This summary table supports regional benchmarking, policy evaluation, and targeted intervention planning by comparing energy and emissions performance across authorities.

6. Average CO2

```
AvgCO2 = AVERAGE(certificate_master[CO2_EMISSIONS_CURRENT])
```

Figure 27. Average CO2 DAX code

To evaluate the current carbon footprint of properties, a DAX measure named AvgCO2 was created. This measure calculates the average of the CO2_EMISSIONS_CURRENT values from the certificate_master table.

Methodological Steps:

- **Function Used:** AVERAGE() computes the mean value of a numeric column, excluding blanks.
- **Source Table:** certificate_master contains energy and emissions data for each property.
- **Column Aggregated:** CO2_EMISSIONS_CURRENT represents the current estimated CO₂ emissions per property.
- **Purpose:** This measure provides a benchmark for assessing the environmental impact of the existing building stock, supporting sustainability analysis and emissions reduction planning.

7. Average Efficiency Measure

```
AvgEfficiencyMeasure = AVERAGE(vw_EnergyRatingDistribution[AvgEfficiency])
```

Figure 28. Average Efficiency Measure DAX code

To support high-level analysis of energy efficiency across grouped property data, a DAX measure named AvgEfficiencyMeasure was created. This measure calculates the average of the AvgEfficiency column from the vw_EnergyRatingDistribution view.

Methodological Steps:

- **Function Used:** AVERAGE() computes the mean of a numeric column, excluding blank entries.
- **Source View:** vw_EnergyRatingDistribution is a structured view that aggregates property-level energy ratings into grouped summaries.

- **Column Aggregated:** AvgEfficiency represents the average current energy efficiency score for each group (e.g., by rating or region).
- **Purpose:** This measure enables streamlined reporting of average efficiency across predefined segments, supporting comparative analysis and performance tracking.

8. Energy Efficiency Gap

```
EnergyGap = certificate_master[POTENTIAL_ENERGY_EFFICIENCY] - certificate_master[CURRENT_ENERGY_EFFICIENCY]
```

Figure 29. Energy Efficiency Gap DAX code

To quantify the potential improvement in energy performance for each property, a variable named EnergyGap was defined. This variable calculates the difference between the potential and current energy efficiency scores from the certificate_master dataset.

Methodological Steps:

- **Operation Used:** Simple arithmetic subtraction to compute the efficiency gap.
- **Source Fields:**
 - POTENTIAL_ENERGY_EFFICIENCY: Projected score if recommended upgrades are implemented.
 - CURRENT_ENERGY_EFFICIENCY: Existing score based on current property conditions.
- **Purpose:** This derived metric highlights the room for improvement in energy efficiency, enabling prioritization of properties with the highest upgrade potential.

9. Average Energy Efficiency Gap

```
AvgEnergyGap = AVERAGE(certificate_master[EnergyGap])
```

Figure 30. Average Energy Efficiency Gap DAX code

To evaluate the typical improvement potential across all properties, a DAX measure named AvgEnergyGap was created. This measure calculates the average of the EnergyGap column from the certificate_master table.

Methodological Steps:

- **Function Used:** AVERAGE() computes the mean of a numeric column, excluding blanks.
- **Source Table:** certificate_master contains both current and potential energy efficiency scores for each property.
- **Column Aggregated:** EnergyGap represents the difference between potential and current energy efficiency, previously derived using Python or DAX.
- **Purpose:** This measure provides a high-level indicator of the average efficiency improvement opportunity across the dataset, useful for policy evaluation, retrofit prioritization, and sustainability reporting.

10. Full Address

```
1 FullAddress =  
2 TRIM(  
3     certificate_master[ADDRESS1] & " " &  
4     certificate_master[ADDRESS2] & " " &  
5     certificate_master[ADDRESS3]  
6 )
```

Figure 31. Full Address DAX code

To generate a complete and readable address string for each property, a calculated column named FullAddress was created using DAX. This column concatenates three separate address fields from the certificate_master table and removes any leading or trailing spaces.

Methodological Steps:

- **Function Used:** TRIM() ensures the final string has no extra spaces at the beginning or end.
- **Concatenation Logic:** The ampersand (&) operator joins ADDRESS1, ADDRESS2, and ADDRESS3 with spaces in between.
- **Source Table:** certificate_master contains segmented address fields for each property.
- **Purpose:** This transformation simplifies address handling for display, filtering, and geolocation tasks by consolidating fragmented address components into a single field.

11. Percentage of Properties Needing Efficiency Improvement

```
PercentNeedsImprovement =  
DIVIDE(  
    COUNTROWS(FILTER(certificate_master, certificate_master[CURRENT_ENERGY_RATING] IN {"D","E","F","G"})),  
    COUNTROWS(certificate_master),  
    0  
)
```

Figure 32. Percentage of Properties Needing Efficiency Improvement DAX code

To quantify the proportion of properties with suboptimal energy ratings, a DAX measure named PercentNeedsImprovement was created. This measure calculates the percentage of entries in the certificate_master table whose CURRENT_ENERGY_RATING falls within the lower efficiency bands: D, E, F, or G.

Methodological Steps:

- **Function Used:** DIVIDE() safely performs division, returning 0 if the denominator is zero.
- **Numerator Logic:** COUNTROWS(FILTER(...)) counts only those records with ratings D to G.
- **Denominator Logic:** COUNTROWS(certificate_master) gives the total number of records.
- **Purpose:** This measure expresses the share of properties that require energy efficiency upgrades, supporting prioritization and policy targeting in sustainability planning.

12. Total Property Count

```
PropertyCountMeasure = SUM(vw_EnergyRatingDistribution[PropertyCount])
```

Figure 33. Total Property Count DAX code

To support analysis of property distribution across energy rating categories, a DAX measure named PropertyCountMeasure was created. This measure calculates the total number of properties recorded in the vw_EnergyRatingDistribution view.

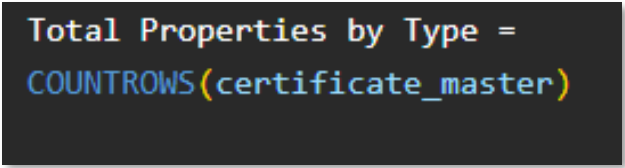
Methodological Steps:

- **Function Used:** SUM() aggregates numeric values across all rows.

- **Source View:** vw_EnergyRatingDistribution is a structured view that summarizes property counts by energy rating.
- **Column Aggregated:** PropertyCount represents the number of properties within each rating group.

Purpose: This measure enables dynamic reporting of total property counts, especially when filtered by rating bands, regions, or inspection years. It supports visualizations such as bar charts, pie charts, and KPI cards.

13. Total Property Count by Type



```
Total Properties by Type =  
COUNTROWS(certificate_master)
```

Figure 34. Total Property Count by Type DAX code

To quantify the total number of property records available for analysis, a DAX measure named Total Properties by Type was created. This measure counts all rows in the certificate_master table, which represents individual property entries.

Methodological Steps:

- **Function Used:** COUNTROWS() returns the total number of rows in a table.
- **Source Table:** certificate_master contains detailed energy certificate data for each property.
- **Purpose:** This measure provides a foundational count of properties, useful for calculating percentages, benchmarking, and validating dataset completeness. It can also be filtered dynamically by property type, region, or inspection year to support segmented analysis.

14. Total Property Count Variable

```
TotalProperties = COUNTROWS(certificate_master)
```

Figure 35.Total Property Count Variable DAX code

To establish a baseline count of all property records in the dataset, a DAX variable named TotalProperties was defined. This variable calculates the total number of rows in the certificate_master table.

Methodological Steps:

- **Function Used:** COUNTROWS() returns the total number of entries in a specified table.
- **Source Table:** certificate_master contains individual property-level energy certificate data.
- **Purpose:** This variable serves as a foundational metric for calculating proportions, validating data completeness, and supporting conditional logic in other measures or calculated columns.

15. Local Authority Summary

```
LocalAuthoritySummary =  
SUMMARIZE(  
    certificate_master,  
    certificate_master[LOCAL_AUTHORITY],  
    "TotalProperties", COUNTROWS(certificate_master),  
    "AvgEfficiency", AVERAGE(certificate_master[CURRENT_ENERGY_EFFICIENCY]),  
    "AvgPotentialEfficiency", AVERAGE(certificate_master[POTENTIAL_ENERGY_EFFICIENCY]),  
    "AvgCO2Current", AVERAGE(certificate_master[CO2_EMISSIONS_CURRENT]),  
    "AvgCO2Potential", AVERAGE(certificate_master[CO2_EMISSIONS_POTENTIAL])  
)
```

Figure 36.Local Authority Summary DAX code

To enable comparative analysis across administrative regions, a DAX variable named LocalAuthoritySummary was created using the SUMMARIZE() function. This variable constructs a summary table that aggregates key energy and emissions metrics for each local authority.

Methodological Steps:

- **Function Used:** SUMMARIZE() groups data by a specified column and allows calculation of multiple aggregated metrics.
- **Grouping Column:** LOCAL_AUTHORITY enables regional breakdown of performance indicators.

- **Calculated Metrics:**

- "TotalProperties": Total number of properties per authority.
- "AvgEfficiency": Average current energy efficiency score.
- "AvgPotentialEfficiency": Average projected efficiency score post-upgrades.
- "AvgCO2Current": Average current CO₂ emissions.
- "AvgCO2Potential": Average potential CO₂ emissions after improvements.

- **Purpose:** This summary supports regional benchmarking, policy evaluation, and targeted sustainability planning by comparing energy and emissions performance across local authorities.

5. Results and Visual Analysis

Following the cleansing and transformation of the data in SQL, the data was brought into Power BI via the SQL Server database connector. The goal was to create a dynamic and easy-to-use dashboard for various stakeholders (i.e., policymakers, analysts, etc.) with different degrees of technical proficiency.

The dashboard design you developed-and this report is solely based upon-includes:

- KPI Cards (Total Properties, Avg Potential Efficiency Score)
- EPC Overview Text Card
- Pie Chart: Distribution of Current Energy Ratings
- Treemap: Distribution of Properties by Energy Rating
- Table Matrix: Energy Efficiency and Improvement Potential by Property Type
- Bar Chart: Average Energy Gap by Property Type, Average CO₂
- Line/Area Chart: Average CO₂ Emissions Over Time
- Map Visualization: Geographic Distribution of CO₂ Emissions
- Slicers: Property Type & Year
- Manchester Flag Graphic

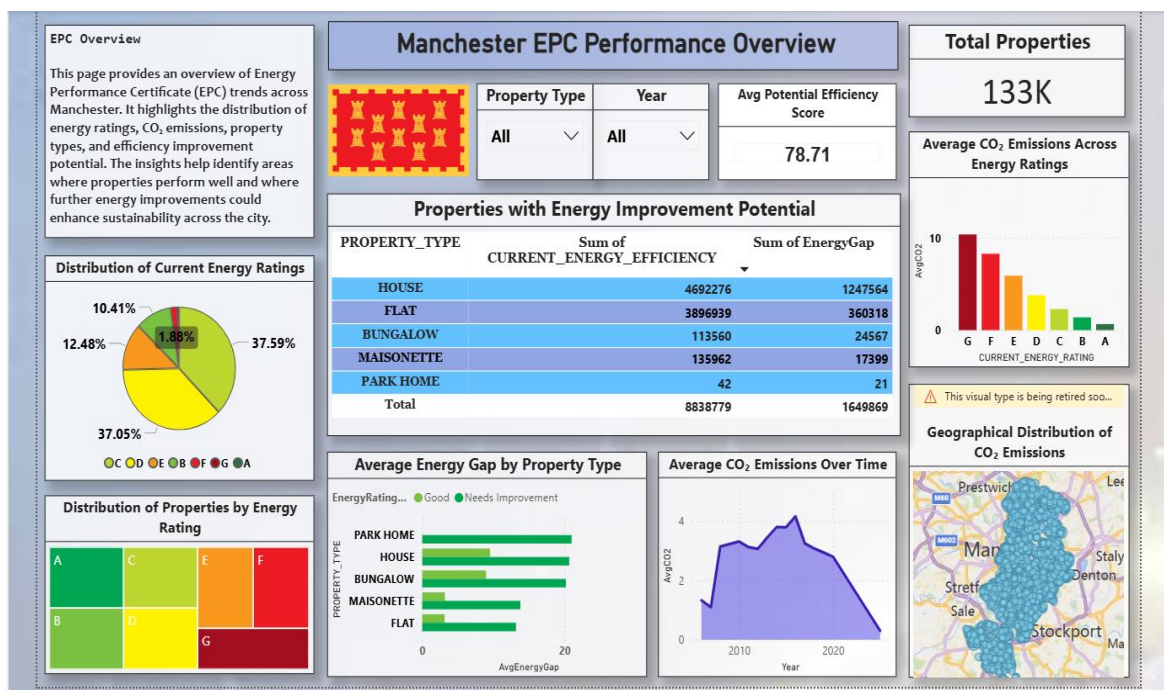


Figure 37.Dashboard View

a. KPI Cards



Figure 38.KPI Cards

What it shows:

- This card displays the total number of properties included in the EPC dataset.

How to create:

1. Go to Visualizations → select Card
2. Add the sum of 'LocalAuthoritySummary (TotalProperties)' to Fields
3. Click the dropdown → select Count
4. Rename title to Total Properties
5. Go to Visualizations again → select Card
6. Add 'certificate_muster (AvgPotentialEfficiency)' to Fields
7. Change aggregation to Average
8. Rename title to Avg Potential Efficiency Score

Total Properties: 133K

- The dataset includes over 133,000 EPC certificates for Manchester, providing strong analytical coverage.

Average Potential Efficiency Score: 78.71

- This suggests a significant opportunity to improve.
- Scores nearing 80 indicate high C or low B performance on the EPC scale.

Interpretation:

- Manchester households are not operating with optimum efficiency.
- The majority of the properties could easily improve with minor investment into measures.

b. Slicers – Property Type and Year

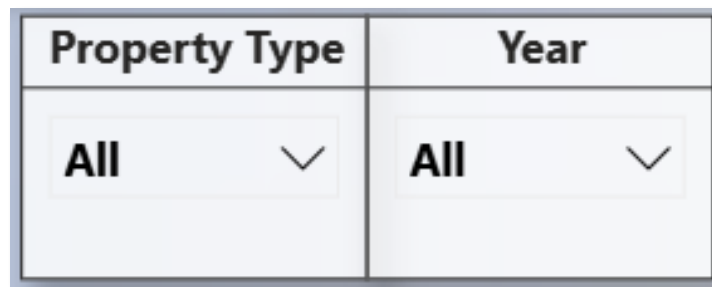


Figure 39. Slicers – Property Type and Year

What it shows:

- Allows users to filter the entire dashboard by property type and Lodgement Year.

How to create:

1. Select Slicer
2. Add Property Type to Field
3. Format → Choose Dropdown
4. Select slicer again
5. Add Lodgement year to the field
6. Format → Choose Dropdown

These slicers allow stakeholders to:

- Filter dashboard visuals dynamically
- Compare performance trends across years
- Focus on specific property types like Houses vs Flats

This dramatically increases dashboard usability.

c. EPC Overview Narrative Box

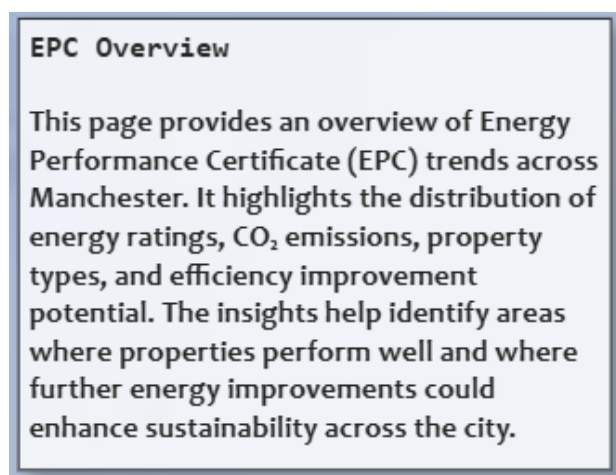


Figure 40. EPC Overview Narrative Box

Placed at the upper-left corner, this text box summarises what the dashboard covers, framing the analysis for stakeholders.

It includes:

- Distribution of ratings
- CO₂ emissions trends
- Property types
- Improvement potential
- Regional variations

This makes the dashboard self-explanatory.

d. Pie Chart – Distribution of Current Energy Ratings

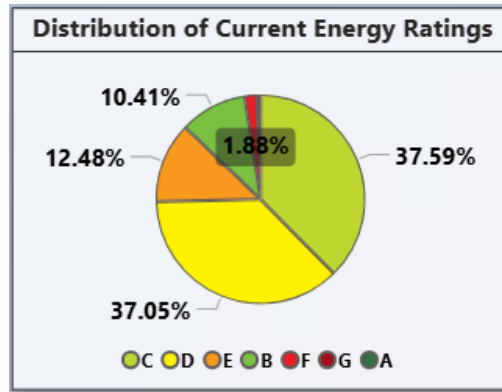


Figure 41. Pie Chart – Distribution of Current Energy Ratings

What it shows:

- The pie chart shows how many properties fall into each EPC rating (A–G).

How to create:

- Go to Visualizations → Pie Chart
- Add 'vw_EnergyRatingDistribution (Current_Energy_Rating)' to Legend
- Add certificate_master (TotalProperties) to Values
- Turn on Data labels
- Use EPC colour scheme (A = green → G = red)

Pie chart includes these **exact proportions**:

EPC Rating	Percentage	Interpretation
C	37.59%	Dominant rating – most houses are in mid-energy band
D	37.05%	Nearly equal to C – indicates mass mid-level performance
E	12.48%	Many older homes fall here
B	10.41%	High-performing homes but still relatively few
A	1.88%	Very energy-efficient homes remain extremely scarce
F / G	~0%	Very few properties in the lowest bands

Key Insight:

- Manchester is overwhelmingly **mid-rated (C–D)** with considerable room for improvement.

e. Treemap – Distribution of Properties by Rating



Figure 42. Treemap – Distribution of Properties by Rating

What it shows:

A treemap showing the size of each EPC rating group, colour-coded.
Larger blocks = more properties in that rating.

How to create:

1. Select Treemap
2. Add 'certificate_master (Current_Energy_Rating)' to Category
3. Add 'certificate_muster (PropertyCountMeasure)' to Values
4. Apply EPC colours (green → red)

This treemap visualizes rating categories using color-coded blocks:

Rating	Meaning	Usual EPC Color
A	Most efficient	Dark Green
B	Very efficient	Green
C	Quite efficient	Light Green
D	Average	Yellow
E	Below average	Orange
F	Inefficient	Red-Orange
G	Very inefficient	Red

Figure 43.color-coded blocks table

This reinforces the pie chart:

- **C and D ratings are the largest blocks**, meaning:
 - Most properties in Manchester fall into **average** or **slightly above-average** energy performance.
- **A and B ratings are very small**, which means:
 - Only a small number of properties are highly energy efficient.
- **E, F, and G have medium-to-small blocks**, meaning:
 - A noticeable portion of homes still have **below-average efficiency**, but fewer than C or D.

Interpretation:

Most homes in Manchester are rated C or D. Very few are rated A or B, and some properties still have low ratings like E, F, or G, meaning there is room for energy improvements.

f. Table – Properties with Energy Improvement Potential

Properties with Energy Improvement Potential		
PROPERTY_TYPE	Sum of CURRENT_ENERGY_EFFICIENCY	Sum of EnergyGap
...		▼
HOUSE	4692276	1247564
FLAT	3896939	360318
BUNGALOW	113560	24567
MAISONETTE	135962	17399
PARK HOME	42	21
Total	8838779	1649869

Figure 44. Table – Properties with Energy Improvement Potential

What it shows:

- Shows the sum or average of current efficiency and total of Energy Gap across each property type.
Good for comparing multiple metrics at once.

How to create:

- Select Matrix visual
- Add 'certificate_master (Property_Type) to Rows
- Add the sum of 'certificate_master (Current_Energy_Efficiency) and the 'sum of certificate_master (EnergyGap) to Values
- Turn on Column subtotals and row subtotals for clarity

Interpretation:

- Most of the occupancy is Houses because Manchester has a very high stock of terraced and semi-detached houses.
- Flats have the second largest occupancy because there are several large blocks of flats at high density.
- Houses represent the greatest potential for improvement, which indicates that targeted policies for this housing type will provide the largest overall city-wide gain.
- The small amount of status for Park Homes is largely because there are not many Park Homes in Manchester, so there is bias in the data.

Key Insight:

- If stakeholders want the greatest overall improvement efficiency for the city on limited budget, Houses will produce the greatest improvement efficiency for the city if up modified or upgraded.

g. Bar Chart – Average Energy Gap by Property Type

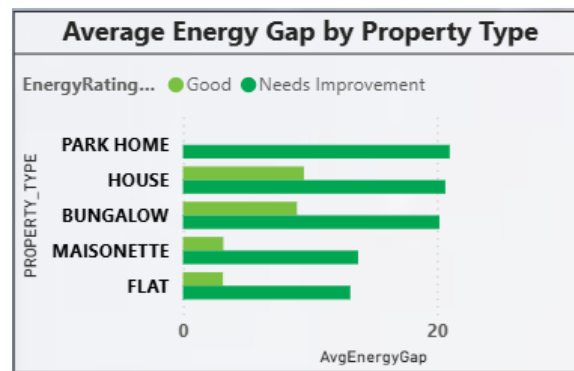


Figure 45. Bar Chart – Average Energy Gap by Property Type

What it shows:

- The chart displays the average improvement potential for each property type.

How to create:

1. Select Clustered Bar Chart
2. Add 'certificate_master (Property_Type)' to Y-axis
3. Add 'certificate_master (AvgEnergyGap)' to X-axis
4. Add 'certificate_master (EnergyRatingStatus)' to Legend

Visual Features:

- Green bars: “Good” (small gap)
- Light green bars: “Needs Improvement” (large gap)

Chart shows:

- **Park Home** – Large improvement potential
- **House** – Substantial gap
- **Bungalow** – Meaningful gap
- **Maisonette** – Moderate gap
- **Flat** – Lowest gap

Interpretation:

Flats often have:

- better insulation due to shared walls
- smaller surface area exposed to heat loss

Houses, especially older terraced houses, tend to:

- lose heat faster
- have outdated insulation

Thus, they show larger gaps.

h. CO₂ Trend Line – Average CO₂ Emissions Over Time

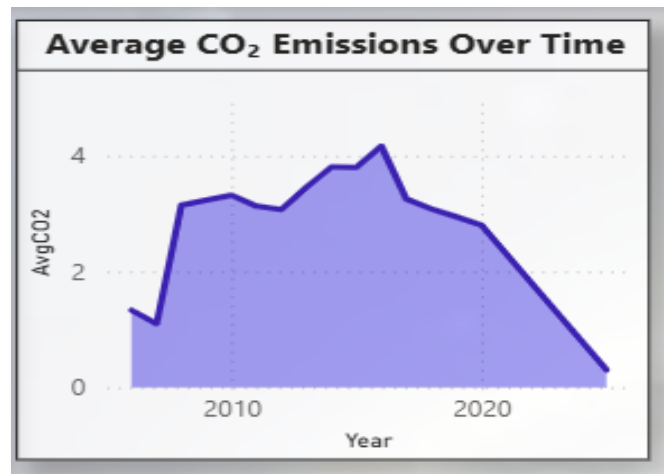


Figure 46.CO₂ Trend Line – Average CO₂ Emissions Over Time

What it shows:

- Displays the trend of CO₂ emissions from 2010 to 2024. Shows whether emissions are rising or falling over time.

How to create:

1. Select Line Chart
2. Add 'certificate_master (INSPECTION_DATE)' and 'Year' to X-axis
3. Add 'certificate_master (AvgCO₂)' to Y-axis
4. Turn on Shade area

Area chart demonstrates:

- Significant fluctuation year-to-year
- General decline from 2014 to 2024
- Peak around ~2016–2018
- A downward slope after 2020

Interpretation:

This suggests:

- Improvements in heating systems
- Retrofitting schemes (e.g., ECO, Green Homes Grant)
- Behavioural changes
- Decline in reliance on older boilers

Even though variation exists, the long-term emission reduction trend is a positive indicator.

i. Map Visual – Geographic Distribution of CO₂ Emissions

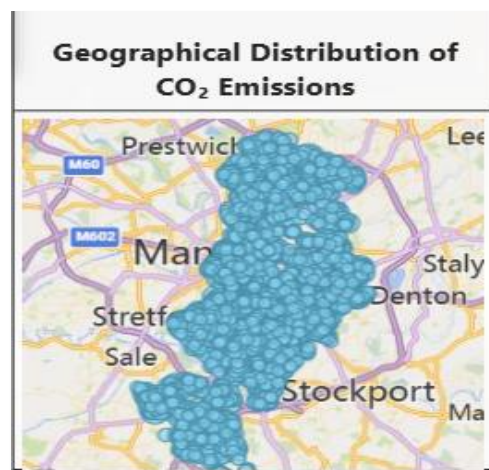


Figure 47. Map Visual

What it shows:

- Plots each property on the map using latitude and longitude, sized or coloured by CO₂ emissions. Shows high-emission clusters geographically.

How to create:

1. Select Map visual
2. Add Postcode to Location
3. Add the 'sum of certificate_master (CO2_EMISSION_CURRENT)' to Bubble size
4. Add First 'certificate_master (CURRENT_ENERGY_RATING, LOCAL_AUTHORITY, PROPERTY_TYPE, EnergyGap)' to Tooltips.

This visual plots properties by geographic coordinates.

Map Insights:

- Dense clusters appear in the city centre and inner Manchester
- Outer districts have fewer properties, shown by fewer points
- Variation in CO₂ values is visible by point colour intensity (depending on your map theme)

Interpretation:

The densest clusters align with:

- High population density
- Large numbers of flats and older buildings
- Areas where energy performance tends to vary widely

16. Discussion

a. Housing Characteristics

Most properties in Manchester are older homes, which results in a majority of properties achieving EPC ratings of C and D and having little to no newer energy efficient properties. Older homes generally show the greatest opportunity for improvements to insulation and heating systems throughout the city.

b. Property-Type Differences

The efficiency gaps exist predominantly within Detached Houses, whereas the differences in efficiency within Flats is much smaller due to their proximity to other Flats, therefore confirming that property types must be taken into consideration when designing strategies for retrofitting existing homes. The primary target area for retrofitting should be Detached Houses.

c. CO₂ Emissions Trend

CO₂ emissions are forecast to continue decreasing between 2014 and 2024; however, there will be fluctuations in the velocity of the reductions. Continued support through successful policy development and implementation, continued retrofit funding, and investments in energy efficient systems are all critical for achieving reductions.

d. Geographical Patterns

Higher rates of CO₂ emissions are being identified as trends in Central or Older Neighbourhoods, which raises concerns about spatial energy inequality throughout the region. Emphasis should be placed on developing targeted retrofit projects in these locations.

17. Recommendations

a. Local Authority Retrofit Priorities

The primary focus of retrofitting for local authorities should be in areas where houses make up the majority of the buildings and have high levels of emissions and gaps in efficiency. The most important types of retrofits that should be upgraded include insulation, window glazing and improvements to heating systems.

b. Role of Energy Providers

Energy suppliers can use CO₂ hotspot mapping as a tool to inform the type of advisory services they can provide and to work alongside retrofit initiatives to help drive consumer engagement and meet CO₂ reduction targets.

c. Future Research Directions

Integrating EPC data with socio-economic, environmental and building-use datasets would enable the development of predictive modelling technologies and the creation of more accurate estimates of energy-related risk at a spatial level.

d. Policy Enhancements

Enhanced regulation of low EPC rated properties particularly within the rental market is required through regulatory mechanisms and enhanced financial incentives as well as increased frequency of EPC assessments to ensure compliance and data accuracy are improved.

e. Guidance for Landlords and Homeowners

Property owners should implement low-cost improvement actions to reduce energy usage e.g. draught proofing, installation of LED light bulbs, and upgrade of loft insulation. In addition, properties which are classified as 'houses' would achieve greater energy efficiency by upgrading their central heating systems and upgrading their wall insulation.

18. Limitations

a. Data Quality and Completeness

In the original dataset, several entries were missing or invalid (e.g., CO₂ emissions and energy scores, which even at times were negative), so they were also cleaned. We undertook a systematic clean-up process in SQL, which poses a slight risk of distorting the dataset. Moreover, the dataset only contains properties assessed recently, which means that older buildings—or buildings that have not been assessed—are not represented in our study.

b. Limitations of EPC-based Measures

EPC ratings represent anticipated performance under standardised conditions, rather than actual energy consumption. Consequently, factors in the real world—occupant behaviour, warming preferences, appliance efficiency, etc.—would not be recorded. Thus, insights from EPC-based metrics may not always as promissably correlate with real-world household energy use.

c. Temporal and Policy Changes

EPC methodology has changed between 2014 and 2024. Changes in the scoring formula, inspection procedure or government regulations, are all other factors that may have causal factors in measured trends. This means part of the variation over time might be influenced by changes in the EPC, not through genuine improvements in the energy performance of the property assessments.

19. Conclusion

This project has offered a comprehensive analytic evaluation of residential energy performance in Manchester using the Energy Performance Certificate (EPC) dataset, SQL Server, and Power BI. By structured data extraction, scrupulous cleaning in SQL, and a complex interactive dashboard, we were able to analyse a large amount of EPC records and convert them to usable, informative, and robust ideas.

The analysis shows properties within Manchester are predominantly in the mid-range EPC ratings of C and D implying appliances and insulation giving moderate efficiency, with lots of opportunity to improve. Houses provide the biggest share of overall potential improvements, while flats have generally better performance, however their numbers present a level of challenge that requires investigation. CO₂ emissions have steadily reduced over the past ten years, which can be attributed to continual improvements in insulation standards and heating appliances and technologies. The geographical trends produced within the map visual also demonstrate grouping/repeated clustering higher rates of emission properties which demonstrates where specific retrofit programs could create the biggest influence.

The relationships generated between energy efficiency, CO₂ emissions, property type, and area/location demonstrate the need for targeted, planned-based, and data focused intervention. Lower efficiency ratings were accompanied by higher emissions, and specific neighbourhoods consistently performed worse than others. They can be valuable influences for larger and greater systems of policy, local authority, housing provider practitioners or alike, all aimed at decreasing levels of emissions and bettering standards of living.

Future research could integrate socio-economic indicators combined, model retrofit scenarios, or utilize predictive analytics to prioritize the highest risk properties through a vulnerability assessment. In general, this study makes a clear, evidence-based contribution to our understanding of energy efficiency in Manchester and lends support to existing initiatives aiming a more sustainable and energy-efficient housing sector.